

Measurement of the latent heat of solidification of H₂O.

Ben Thomas, Josué Eduardo Tello Guizado

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Abstract

The objective of this work is the determination of the heat of solidification and fusion of H₂O using the calorimeter B calibrated in practice 4. The latent heat of solidification of H₂O determined is $L_{solidification} = -440 \pm 70$ J/g and the latent heat of fusion is $L_{fusion} = 415 \pm 70$ J/g which are compatible with the reference value $L=334$ J/g.

1 Introduction

The determination of the latent heat of water is done using the differential calorimeter used in practice 4. The experiment consists of adding water to one of the containers and cooling the calorimeter by adding liquid nitrogen so that the water reaches the solidification temperature. The latent heat of fusion of water can also be determined by a heating process with a dryer. Using the data program, the calorimetric signal as a function of time is recorded, allowing the calculation of the heat exchanged during the solidification phase, which corresponds to the area under the curve.

2 Methods

Using the calorimeter utilized in the laboratory, the heat flux $\dot{q}$ is obtained by measuring a potential difference that is generated in the presence of a heat increment ΔQ in the system and calibrating the device using a sensitivity curve as a function of temperature $S(T)$. Since this varies with time, $S(T) = S(T(t)) = S(t)$, it can be written as:

$$\dot{q}(t, T) = \frac{V(t)}{S(T)} = \dot{q}(t) = \frac{V(t)}{S(t)} = \frac{V_m(t) - V_b(t)}{S(t)} \quad (1)$$

where $V_m(t)$ is the measured voltage and $V_b(t)$ corresponds to the baseline correction curve. This quantity allows the measurement of the total heat Q that a system absorbs or releases, expressed in Equation 2.

$$Q = \int \dot{q}(t) dt = \int \frac{V_m(t) - V_b(t)}{S(t)} dt \quad (2)$$

It is unknown how the temperature varies as a function of time during the process, but due to its rapidity, it can be assumed to be invariant, constant at the temperature at which it solidifies. Integrating over the time period of the process yields the released heat (Q_c) during this period, and, using the mass M , the latent heat of solidification L is expressed as:

$$Q_c = \frac{1}{S(T_s)} \int_{t_i}^{t_f} (V_m(t) - V_b(t)) dt \implies L = \frac{Q_c}{M} = \frac{1}{M \cdot S(T_s)} \int_{t_i}^{t_f} (V_m(t) - V_b(t)) dt \quad (3)$$

Since discrete values of $V_m(t)$ are obtained in the laboratory, the integral in Equation 3 must be discretized or the function must be approximated using a mathematical fit. The second case has been chosen.

3 Propagation of Uncertainties

To obtain the error committed in the calculation of L , the error propagation law shown in Equation 4 is used.

$$\delta F(x_1, x_2, \dots, x_n) = \sqrt{\sum_i^n \left(\delta x_i \frac{\partial F}{\partial x_i} \right)^2} \quad (4)$$

Since $L = L(M, Q)$, it can be written as

$$\delta L = \sqrt{\left(\delta M \frac{\partial L}{\partial M} \right)^2 + \left(\delta Q \frac{\partial L}{\partial Q} \right)^2} = \sqrt{\left(\frac{\delta M}{M} \right)^2 + \left(\frac{\delta Q}{Q} \right)^2}$$

Considering the expression for Q , to facilitate the calculation of its uncertainty, the integral is approximated by a sum, so the total uncertainty δQ is the sum of the partial uncertainties $\delta(q_i \Delta t_i)$:

$$Q = \int \dot{q}(t) dt \implies Q \approx \sum_i^n \dot{q}(t_i) \Delta t_i \implies \delta Q = \sum_i^n \left(\delta(\dot{q}(t_i) \Delta t_i) \right)$$

$$\delta Q = \sum_i^n \left[\sqrt{\left(\delta V_i \frac{\partial \dot{q} \Delta t_i}{\partial V_i} \right)^2 + \left(\delta S_i \frac{\partial \dot{q} \Delta t_i}{\partial S_i} \right)^2 + \left(\delta \Delta t_i \frac{\partial \dot{q} \Delta t_i}{\partial \Delta t_i} \right)^2} \right]$$

Neglecting the error in time, assuming S invariant in t and knowing that the samples are taken at approximately equal intervals $\Delta t_i = \Delta t_j$:

$$\delta Q = \sum_i^n \left[\sqrt{\left(\delta V_i \frac{\Delta t_i}{S_i} \right)^2 + \left(\delta S_i \frac{V_i \Delta t_i}{S_i^2} \right)^2} \right] = \delta Q = \frac{\Delta t}{S} \sum_i^n \sqrt{\left(\delta V^2 + \frac{\delta S^2}{S^2} V_i^2 \right)}$$

Thus, the final error δL is described by Equation 5.

$$\delta L = \sqrt{\left(\frac{\delta M}{M} \right)^2 + \left(\frac{\Delta t}{S \cdot M} \sum_i^n \sqrt{\left(\delta V^2 + \frac{\delta S^2}{S^2} V_i^2 \right)} \right)^2} \quad (5)$$

4 Results

The tendency of the recorded calorimetric signal data as a function of time for the cooling case is shown in Figure 1, where a peak is observed at a temperature of -14.1°C due to the phase change by solidification. The data related to the heating process is shown in Figure 2, where a peak is observed at a temperature of 8.2°C due to the phase change, in this case, it is melting.

The temperature of solidification and melting of water differs from the theoretical values because water is in metastable states, and therefore, any slight disturbance will cause water to change states and reach an equilibrium state by changing phase.

The latent heat of solidification and melting are calculated as the area under the curve and are determined using Equation 2, considering that the mass of the crucible is $m_{\text{crucible}} = 1.96 \pm 0.01$ g and the mass of water is $m_{\text{water}} = 0.05 \pm 0.01$ g. Sensitivity is determined by using the equation 6 which has been modified due to mistakes related to intensity units. The correction of the sensitivity curve is essential because the results of the latent heat differs from the theoretical value one order of magnitude.

$$S(T) = 7.57 \cdot 10^{-4} T - 0.092 \quad (6)$$

In the case of cooling, the initial time is $t_i = 745$ s. By intersecting the baseline correction and a high-order polynomial fit $P(t)$ ($R^2 = 0.9994$) to the curve of the experimental values in the solidification domain, the final time of the solidification process $t_f = 1080$ s is obtained. Using a mass $M = 5 \cdot 10^{-5}$ Kg, and a sensitivity $S(-14.1^\circ\text{C}) = S(259.05\text{ K}) = 1.041$, we obtain:

$$L_{\text{solidification}} \approx \frac{1}{M \cdot S(T_s)} \int_{754}^{1080} \left(P(t) - V_b(t) \right) dt = -440 \pm 70 \text{ J/g}$$

It can be seen that this value, however, is not compatible with the value from literature $L_{solidification} = 334 \text{ J/g}$. After searching and investigating the possible mistake, it has been observed an error in the calibration of practice 4, leading to an incorrect $S(T)$. The latter has been corrected and computed once more, leading to a new value of $S(259.02) = 0.1041$, and reaching a final result of:

$$L_{solidification} = -440 \pm 70 \text{ J/g}$$

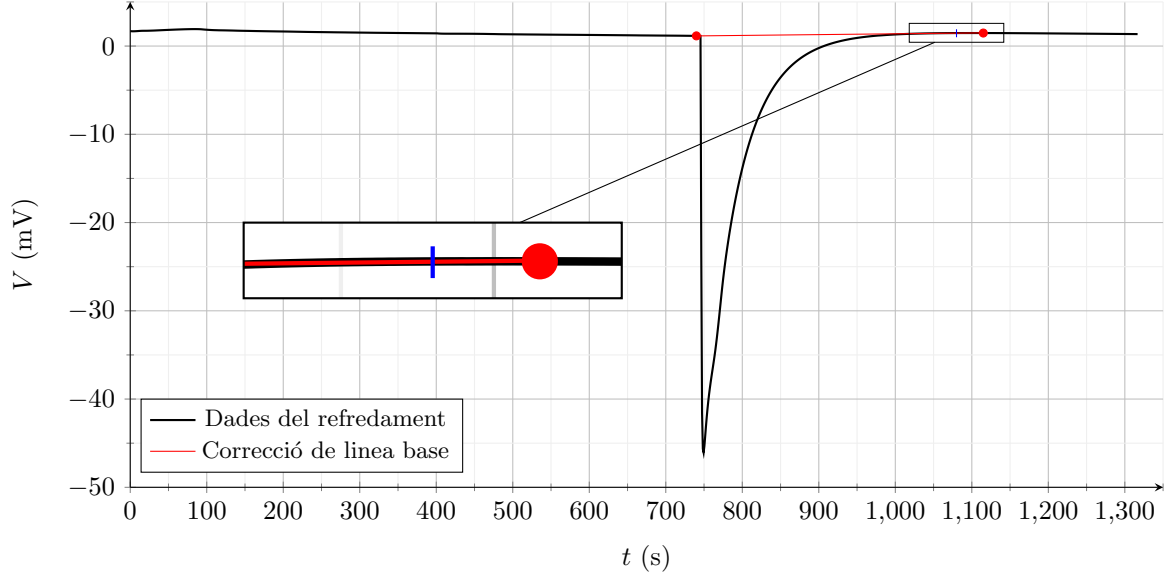


Figure 1: Voltatge V detectat en funció del temps t per al refredament de l' H_2O

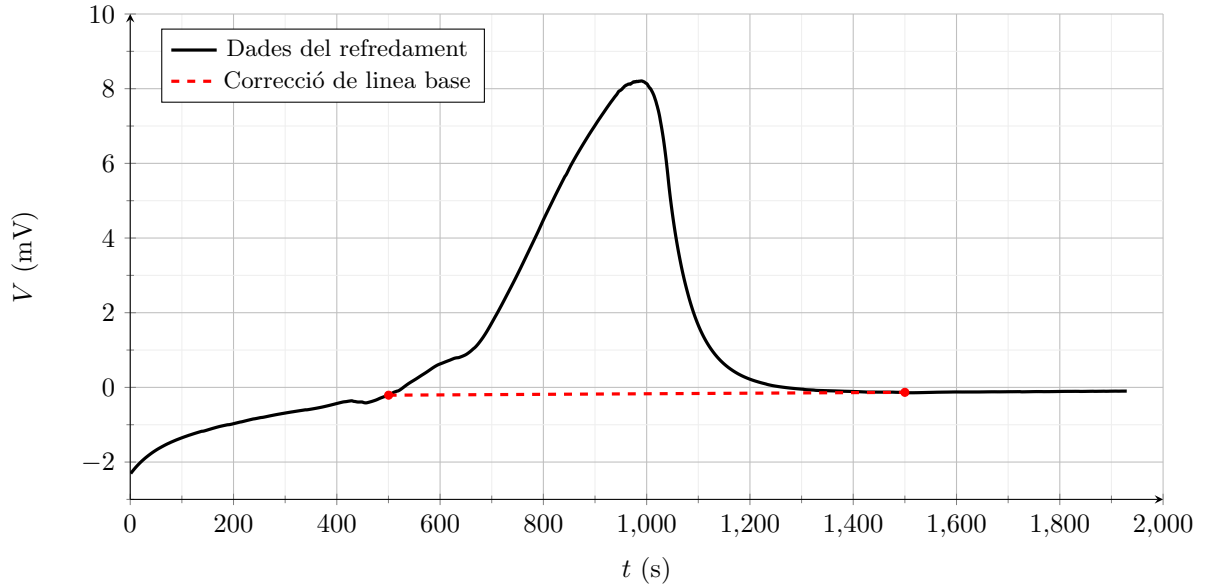


Figure 2: Voltatge V detectat en funció del temps t per al refredament de l' H_2O

In the case of heating, the initial time is $t_i = 500.32 \text{ s}$. By intersecting the baseline correction and a high-order polynomial fit $P(t)$ ($R^2 = 0.97$) to the curve of the experimental values in the fusion domain, the final time of the fusion process $t_f = 1500.29 \text{ s}$ is obtained. Using a mass $M = 5 \cdot 10^{-5}$

Kg, and a sensitivity $S(8.2^\circ C) = S(281.35 K) = 0.121$, we obtain:

$$L_{fusion} \approx \frac{1}{M \cdot S(T_f)} \int_{500.32}^{1500.29} \left(P(t) - V_b(t) \right) dt = 415 \pm 70 \text{ J/g}$$

$L_{fusion} = 415 \pm 70 \text{ J/g}$

5 Conclusions

The determined latent heat of solidification of water is $L_{solidification} = -440 \pm 70 \text{ J/g}$, and for melting, $L_{fusion} = 415 \pm 70 \text{ J/g}$. The experimental values are compatible with the reference value $L=334 \text{ J/g}$ as the discrepancy $d = 106 J/g < 2\sigma_{solidification} = 140 J/g$ for solidification and $d = 81 J/g < 2\sigma_{fusion} = 140 J/g$. for the fusion process.